

Summer term 2026

Visual Data Analysis

Assignment Sheet 4

Solution has to be submitted via eCampus by May 8, 2026, 10:00 a.m.

If you have questions concerning the exercises, please use the forum on eCampus.

- Please work on this exercise in **small groups** of 3 students. Submit each solution only once, but clearly indicate who contributed to it by forming a team in eCampus. Remember that all team members have to be able to explain all answers.
- Please submit your answers in PDF format, and your scripts as *.py/*.ipynb files. If you are using [Jupyter notebook](#), please also export your scripts and results as PDF.

Exercise 1 (Interactive Exploration of Parameters in ISOMAP, 16+4 Bonus Points)

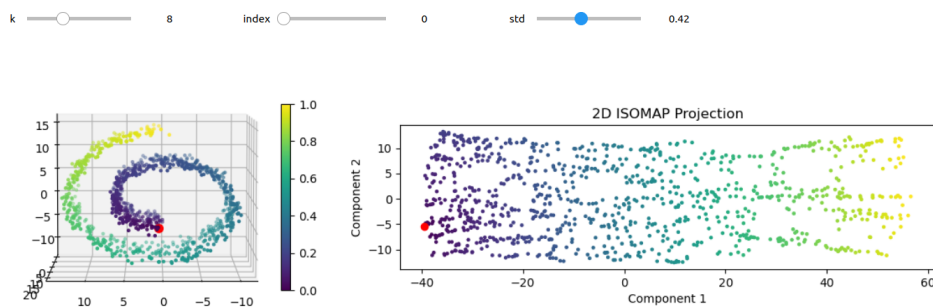


Figure 1: 3D swiss roll data and its 2D ISOMAP projection.

In this task, you will explore the role of the neighborhood size and the level of noise in the data on ISOMAP embeddings, based on the famous swiss roll dataset. You will also learn to use IPyWidgets to interactively set the parameters of a visualization in a Jupyter notebook.

- sklearn contains a function to generate the swiss roll data that was introduced in the lecture. Use $n = 1000$ sample points, a standard deviation of $\text{std} = 0.42$ and random seed 42. Visualize the swiss roll in a 3D plot with a viewing angle that roughly matches the one in Figure 1. (2P)
- Apply ISOMAP with a suitable number k of neighbors to un-roll the data in two dimensions. Choose the left-most point w.r.t. the first component of the embedding as a reference and identify its row in the corresponding dissimilarity matrix. Color each point in both the 2D embedding and the 3D plot based on its geodesic distance from the representative, using a color gradient to indicate relative proximity. (4P)
- [IPyWidgets](#) are a general purpose tool-kit to build interactive controls in Jupyter notebooks. In this subtask, you will use them to make your data generator interactive. Initialize a slider for the standard deviation in the swiss roll data generator, the number of neighbors for building the neighborhood graph in the ISOMAP algorithm, and an index over all data points, to set the reference for color coding distances. Make sure all sliders operate in a sensible value range. (2P)

- d) Write a function that wraps your visualization from the previous task b) with the slider values as inputs. Instead of always using the left-most point in the embedding as a reference for the coloring, let the index select the candidate based on an ordered list from low to high w.r.t. the first ISOMAP component. Arrange the plots similar to Figure 1. (5P)
- e) Using a fixed number of $k = 8$ neighbors, gradually increase the standard deviation of the noise in the swiss roll data until ISOMAP first fails to reconstruct the true underlying manifold. Submit a screenshot that highlights the problematic region. Submit another screenshot after trying to fix the problem by changing the ISOMAP parameters. (3P)
- f) **Bonus:** Depending on which slider a user adjusts, many of the steps do not need to be re-calculated. Find a suitable caching scheme for intermediate results to make the visualization more responsive for certain inputs. (4P)

Exercise 2 (Dimension Reduction on Mice Protein Expression Dataset, 13 Points)

On Sheet 3, we introduced a mice protein expression dataset (`Data_Cortex_Nuclear.xls`). If you do not remember the details, please review its description there. In the current task, you will visualize the distribution of all protein expression levels via different dimension reduction techniques, and reason about what answers the resulting views suggest to the following questions that one might pose about this dataset. For simplicity, we will focus on the shock-context (SC) condition.

1. When comparing treatment with memantine vs. saline solution, does the treatment have a measurable effect on Ts65Dn mice (t-SC)? Does it also have an effect on wildtype controls (c-SC)?
 2. Do the expression profiles of Ts65Dn mice and wildtype controls differ without effective treatment (saline solution)?
 3. Do the expression profiles of Ts65Dn mice who have been treated with memantine become indistinguishable from those of wildtype controls (either with or without treatment)? If not, do they become more similar?
- a) Read the data, filter out items with missing values, and select the relevant classes c-SC-s, t-SC-s, c-SC-m, and t-SC-m from the remaining ones. How many items remain for each of the four classes? (1P)
 - b) Remove all columns that contain meta information, so that exactly the 77 dimensional protein expression levels remain. Use PCA to reduce this data to two dimensions, and answer the questions above based on your impression from the resulting scatter plot. (3P)
 - c) Produce a corresponding plot with ISOMAP. Try different values for the number of neighbors, and justify your final choice. Comment on whether your answers to the three questions above would differ from those that were based on PCA. (3P)
 - d) Produce a corresponding plot with t-SNE. Try different values for the perplexity, and justify your final choice. Comment on whether your answers to the three questions above would differ from those that were based on ISOMAP. (3P)
 - e) Produce a corresponding plot with LDA. Comment on whether your answers to the three questions above would differ from those that were based on the unsupervised methods. (3P)

Exercise 3 (Laplacian Eigenmaps vs. t-SNE, 9 Points)

Laplacian Eigenmaps and t-SNE share the focus on keeping neighbors together, rather than attempting to preserve global distances between widely separated points. This exercise encourages you to consider the relationship between these two methods more deeply.

- a) When we focus on keeping neighbors together, an immediate risk is that the embedding might collapse into a single point. How is that prevented in Laplacian Eigenmaps? How is it prevented in t-SNE? In t-SNE, you can argue either in terms of the cost function or its gradients. (2P)
- b) Even if we prevent a collapse of the overall embedding, we might still see crowding, in which points within individual clusters are placed too close to each other to visually distinguish them. Which of the two approaches do you expect to more effectively prevent this and why? (2P)
- c) Since Laplacian Eigenmaps rely on the graph being connected, we recommended a k nearest neighbor graph to adapt to varying local densities. Explain why edge weighting can still cause problems in such cases. Also state how that problem is avoided in t-SNE. (2P)
- d) It is desirable for dimensionality reduction to be deterministic, so that the same data will be visualized similarly when analyzed repeatedly. Do you see a reason to prefer one of the two approaches over the other in that respect? (1P)
- e) We argued in the lecture that the embedding axes have a natural ordering in PCA (decreasing in terms of amount of variance captured), but are arbitrary in MDS (cost function is invariant to rigid transformations of the embedding). Make a similar argument about the axes in Laplacian Eigenmaps and t-SNE. (2P)

Exercise 4 (Visualizing Dimensionally Reduced Data, 12 Points)

When visualizing high-dimensional data, we can reduce it either to two dimensions and produce a single scatterplot, to three dimensions and produce a 3D scatterplot, whose interpretation typically requires interactions (such as rotation), or to an even larger number of dimensions, and produce a scatterplot matrix (SPLOM). The paper [sedlmair-scatterplot-dimred-2013.pdf](#), which you can find on eCampus, provides some guidance on this decision, based on evaluating these options in terms of how well they perceptually separate classes in various real and simulated datasets. Please read it and answer the following questions in your own words.

- a) According to the authors, is it generally more promising to generate multiple 2D plots with different dimension reduction techniques to obtain a more comprehensive impression of a high-dimensional dataset, or is it better to retain a larger number of dimensions, and visualize them with a SPLOM? Which figure shows the data that is relevant to this question? Does it suggest any exceptions from the general rule? (3P)
- b) The paper includes the quite drastic statement that “interactive 3D rarely helps but often hurts”. State at least two problems of interactive 3D visualizations in this context. Explain why, in another application context, interactive 3D visualizations (not necessarily 3D scatterplots) might still be effective. Give an example of such a case. (4P)
- c) Why did the authors decide to perform a time-consuming manual coding of class separation rather than use an objective measure such as the distance consistency or distribution consistency that we learned about in the lecture? How did they verify the reliability of their subjective impression of class separation? Based on their results, is it possible to state which combination of dimension reduction and visual encoding worked best for each of the datasets? (3P)
- d) SPLOMs are flexible with respect to the dimension, as long as it does not become too large. How has a suitable number of dimensions been selected in this work? How large were the resulting SPLOMs on average? (2P)

Good Luck!